

What a local encoder cannot see: spectral fibres of the human proteome and the representation of perturbagens

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Abstract

UniPert-G2CP (Li et al., *Cell* 189, 2026) represents genetic perturbagens by their protein sequences and chemical perturbagens by ECFP4 fingerprints, and shows that its protein encoder outperforms composition-type encoders such as pseudo amino acid composition (PseAAC). We look at this comparison from the side of algebraic genomics, i.e. the combinatorics of sequences with prescribed local content. An encoder is *local of order k* if it depends on a sequence only through its multiset of k -mers. Composition, k -mer composition, PseAAC with lag $\lambda \leq k - 1$, and every mean-pooled convolutional encoder of window k are local. So is ECFP, in the molecular-graph sense. A local encoder is constant on the *spectral fibre* of a sequence, the set of sequences with the same k -mer multiset. The size of the fibre is an Eulerian-trail count, which we compute exactly by the BEST theorem after contracting forced arcs, for all 20 431 reviewed human proteins. Three findings follow. (i) *Separation is not resolution*. No two distinct human proteins share a k -mer spectrum for any $k \geq 2$, so every local encoder of order ≥ 2 can separate the reference proteome. Yet the median protein shares its 3-mer spectrum with 2^{239} other sequences and its 4-mer spectrum with 2^{19} . The spectral resolution κ , the least k beyond which the spectrum determines the sequence, has median 5 and exceeds 10 for 854 proteins. (ii) *The unresolved order lives in duplication-driven families*. The largest κ occur in the hominoid-specific segmental-duplication families: NBPF, with median $\kappa = 46$ and up to 951, then GOLGA6, NPIP and POTE. They also occur in keratin-associated proteins, collagens and mucins. KRAB zinc-finger proteins are 3% of the proteome but 41% of the proteins with $\kappa > 10$. (iii) *Explicit collisions*. For 12 human proteins we construct a different sequence, altered at up to 15% of its positions, with the same 25-spectrum. It receives the identical 68-dimensional PseAAC vector (with $\lambda = 24$, the setting of the UniPert benchmark) and the identical output from a window-25 convolutional encoder. On the DNA side, which UniPert names as its limitation, a strand-agnostic encoder is constant on the double-stranded reconstruction set. For bacteriophage ϕ X174 at $k = 10$ that set has 31 925 753 246 212 elements.

Keywords. perturbation modelling, protein language models, pseudo amino acid composition,

ECFP, de Bruijn graph, Eulerian trails, BEST theorem, matrix-tree theorem, tandem repeats, segmental duplication, KRAB zinc-finger proteins, reverse complement.

1 Introduction

UniPert-G2CP [1] is a two-stage framework for perturbation modelling. UniPert places genetic and chemical perturbagens in one embedding space. A compound enters as an ECFP4 fingerprint [2]. A gene enters as its protein: mean-pooled ESM-2 embeddings [3] are refined by a graph network over a Smith–Waterman similarity graph of 19 187 human proteins. G2CP then pretrains a perturbation-effect predictor on CRISPR screens and fine-tunes it on compound screens. In the genetic benchmarks (their Fig. 3B, C), UniPert is compared with ESM, OntoProtein and PseAAC. PseAAC was computed with `propy`, 68-dimensional, with lag $\lambda = 24$; this is Chou’s amphiphilic form [4, 5].

The question we ask is structural: *what can an encoder of a given form not distinguish?* For the family of encoders built from local statistics the answer is exact, and it is a question of algebraic genomics. In earlier work one of us (R.C.) counted the circular sequences that share a double-stranded k -mer spectrum with a genome, by a lattice sum over the circulations of a de Bruijn double cover [15, 16, 17]; that work is collected in the monograph [14], which also contains a version of the present paper as its Chapter 12 (pp. 89–99). For single-stranded linear sequences the same count is a single BEST determinant [6, 7]. It can be computed for every protein of the human proteome, and it measures precisely what a local encoder discards.

Section 2 sets up local encoders and their fibres, and proves the counting formula. Section 3 computes fibres and spectral resolution for the proteome. Section 4 locates the unresolved order in protein families. Section 5 constructs explicit collisions for PseAAC and for convolutional encoders. Section 6 treats ECFP, and Section 7 DNA perturbagens. Section 8 states what the results do and do not imply for UniPert-G2CP.

2 Local encoders and spectral fibres

Let Σ be an alphabet, $S \in \Sigma^L$ and $2 \leq k \leq L$. The k -spectrum $\text{spec}_k(S)$ is the multiset of the $m = L - k + 1$ substrings of length k of S . The *fibre* of S is

$$F_k(S) = \{T \in \Sigma^L : \text{spec}_k(T) = \text{spec}_k(S)\}, \quad N_k(S) = |F_k(S)|.$$

We let $F_1(S)$ be the set of rearrangements of S , so that $N_1(S) = L! / \prod_a n_a!$.

Definition 1. An encoder $\phi: \Sigma^* \rightarrow \mathbb{R}^d$ is *local of order k* if $\phi(T) = \phi(S)$ whenever $\text{spec}_k(T) = \text{spec}_k(S)$ and T, S have the same first $(k - 1)$ -mer.

The first $(k - 1)$ -mer is part of the definition only for bookkeeping. If the first and last $(k - 1)$ -mers of S differ, then every $T \in F_k(S)$ has the same first and last $(k - 1)$ -mers as S , because they are the unique vertices of unequal in- and out-degree below.

Proposition 2 (local encoders). *The following are local of order k :*

1. *composition, and j -mer composition for $j \leq k$;*
2. *Chou’s pseudo amino acid composition, in the original and in the amphiphilic form, with any property scales, any weight and any lag $\lambda \leq k - 1$;*

3. every encoder of the form $\phi(S) = \frac{1}{n} \sum_i f(W_i)$, where $W_1, \dots, W_n$ are the windows of length k of S padded with $k-1$ sentinel symbols at each end and f is arbitrary. This includes every convolutional network of receptive field k followed by mean or sum pooling.

Proof. All three are functions of multisets of short substrings. For (1), the j -mer multiset is the multiset of j -prefixes of the k -mers, together with the j -mers inside the last $(k-1)$ -mer. For (2), PseAAC is a function of the composition and of the correlation factors $\tau_j = \frac{1}{L-j} \sum_i \Theta(S_i, S_{i+j})$, $j \leq \lambda$. Here τ_j depends only on the multiset of pairs (S_i, S_{i+j}) , i.e. on the first and last letters of the $(j+1)$ -mers. For (3), the padded windows are the k -mers of S together with windows that meet the sentinels, and those contain only the first or the last $k-1$ letters. $\square$

Being local is a property of the *form* of an encoder, not of its training. A mean-pooled transformer such as ESM-2 is not local, because attention couples all positions. An alignment score against a reference set is not local either.

Theorem 3 (fibre size). *Let $G_k(S)$ be the multigraph on the $(k-1)$ -mers of S with one arc $a[1..k-1] \rightarrow a[2..k]$ for each occurrence of each k -mer a . Let $\mu(a)$ be its multiplicity, s and t the first and last $(k-1)$ -mers of S , and d^+ the out-degree. If $s \neq t$, add one arc $t \rightarrow s$. Write τ_s for the number of spanning in-arborescences rooted at s , with parallel arcs counted separately. Then*

$$N_k(S) = \frac{m^{[s=t]} \cdot \tau_s \cdot \prod_v (d^+(v) - 1)!}{\prod_a \mu(a)!}, \quad (1)$$

where $[s=t]$ is 1 if $s=t$ and 0 otherwise. The whole numerator is formed before the division.

Proof. Label the arcs. The strings with k -spectrum $\text{spec}_k(S)$ correspond to the Eulerian trails of $G_k(S)$, each string to $\prod_a \mu(a)!$ labelled trails. If $s \neq t$, the trails run from s to t and correspond to the Eulerian circuits of the augmented graph that begin with the added arc. By the BEST theorem [6] there are $\tau_s \prod_v (d^+(v) - 1)!$ of them. If $s = t$, the trails are closed. Every labelled circuit gives m labelled closed trails, one for each starting arc, so the count is $m \tau_s \prod_v (d^+(v) - 1)!$. The factor m must be applied before dividing by $\prod \mu(a)!$. Dividing first does not give an integer, e.g. for $S = \text{AAA}$, $k = 2$. $\square$

Remark 4 (relation to the BEST theorem). For a connected balanced multigraph G with labelled arcs, the BEST theorem counts Eulerian circuits up to rotation: $\text{ec}(G) = \tau_r(G) \prod_v (d^+(v) - 1)!$, the same for every root r [6]. Formula (1) is ec of the closed-up graph with two corrections. The factor $m^{[s=t]}$ restores the starting point that “up to rotation” forgets. The division by $\prod_a \mu(a)!$ removes the labels, because a k -mer occurring μ times gives $\mu!$ labelled copies of the same string. The same constant appears in the count $N_{\text{seq}}(c) = t(G_c) \text{Loc}(G_c) / \prod_a c(a)!$ for circular sequences [15, Theorem 7.6 and Remark 7.7, p. 52].

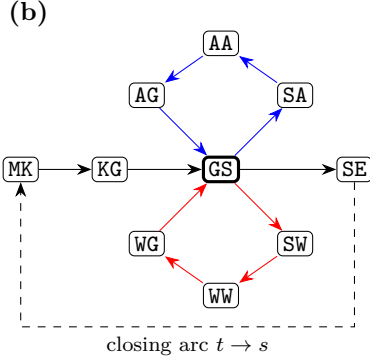
When $s = t$, the fibre also contains the rotations that start elsewhere. The encoders of Proposition 2 are constant on the subset that starts at s . In the human proteome, $s = t$ occurs for a handful of proteins and only at $k \leq 3$.

Lemma 5 (forced arcs [17, Lemma 9.7, p. 71]). *If a vertex $v \neq r$ has all its out-arcs ($w \geq 1$ if them) ending at one vertex $u \neq v$, then $\tau_r(G) = w \tau_r(G/vu)$. Here G/vu merges v into u , deletes the arcs $v \rightarrow u$ and redirects the arcs into v .*

For $k \geq 4$ almost every $(k-1)$ -mer of a protein occurs once, and the lemma reduces τ to the determinant of a small reduced Laplacian. The fibre moves are also classical. Two strings with the

(a) $S = \text{MKGS}\textcolor{blue}{AA}\textcolor{red}{GS}\textcolor{red}{WW}\textcolor{blue}{GSE}$
 $T = \text{MKGS}\textcolor{red}{WW}\textcolor{blue}{GS}\textcolor{blue}{AA}\textcolor{red}{GSE}$

the two segments between the three copies of **GS** change places;
 $\text{spec}_3(S) = \text{spec}_3(T)$, and $N_3(S) = 2$: the fibre is $\{S, T\}$



(c)

local of order 3

composition, dipeptide composition,
PseAAC with $\lambda \leq 2$, window-3 CNN
with mean pooling: $\phi(S) = \phi(T)$

not local

ESM-2 (attention couples all positions),
alignment to a reference graph:
 $\phi(S)$ and $\phi(T)$ may differ

every vertex but **GS** has one out-arc and is contracted;
 $d^+(\text{GS}) = 3$, $\tau = 1$, so $N_3 = \tau(3-1)! = 2$

Figure 1: A spectral fibre in miniature. (a) Two sequences that differ by a transposition of the segments between three copies of the repeated 2-mer **GS**. (b) Their common de Bruijn graph for $k = 3$. The fibre size follows from Theorem 3 after contracting forced arcs (Lemma 5). (c) Every local encoder of order 3 assigns them the same vector. In the human proteome the same mechanism acts on repeat copies 25 residues long and longer (Table 4).

same first $(k-1)$ -mer have the same k -spectrum iff one is obtained from the other by a sequence of transpositions of segments bounded by repeated $(k-1)$ -mers [8, 9]. A local encoder of order k is therefore blind to exactly those rearrangements: swaps of segments that begin and end in repeated $(k-1)$ -mers. Figure 1 shows the smallest interesting case.

Definition 6. The *spectral resolution* of S is $\kappa(S) = \min\{k : N_{k'}(S) = 1 \text{ for all } k' \geq k\}$. It is at most two more than the length of the longest repeated substring of S .

3 The human proteome

We take the 20 431 reviewed human entries of UniProt [10], with 11 418 237 residues, median length 415 and maximum 34 350 (titin). For every protein and every k from 1 up to the point where no $(k-1)$ -mer repeats, we compute N_k by Theorem 3 and Lemma 5. The computation is exact whenever the contracted Laplacian has order at most 150. Larger orders occur only at small k in long proteins, where $\log_2 N_k$ is in the thousands. There the determinant is taken in floating point, and the logarithms of the factorials come from $\log \Gamma$. Against exact values on 200 random sequences, the worst relative error of that path is $6.5 \cdot 10^{-16}$. The formula itself is checked against exhaustive enumeration in 21 187 cases (alphabets of size 2–4, lengths up to 12). The contraction is checked against the full determinant on 400 random sequences. The whole run takes 18 minutes on 16 processes.

Table 1 summarizes the fibres. From $k = 5$ on, the median of $\log_2 N_k$ is 0, i.e. $N_k = 1$: more than half of all human proteins (53.5%) are already determined by their 5-mer spectrum, and the median describes the resolved majority. The tail is what matters, and it is in the second and last columns (Figure 2a). The spectral resolution has median $\kappa = 5$ (Figure 2b). Its 90% and 99%

k	proteins with $N_k > 1$	share	median $\log_2 N_k$	largest $\log_2 N_k$
1	20 431	100.00%	1628.4	140 656.7 (TTN)
2	20 422	99.96%	1080.5	136 673.4 (TTN)
3	20 357	99.64%	239.4	114 168.6 (TTN)
4	18 391	90.02%	18.9	55 107.2 (TTN)
5	9 504	46.52%	0	13 489.5 (TTN)
6	3 445	16.86%	0	6 247.9 (MUC3B)
8	1 346	6.59%	0	3 715.5 (MUC3B)
10	854	4.18%	0	2 252.5 (MUC3B)
12	614	3.01%	0	1 269.2 (MUC3B)
16	327	1.60%	0	404.2 (MUC4)
20	194	0.95%	0	255.4 (MUC2)
25	114	0.56%	0	164.2 (MUC2)
30	81	0.40%	0	108.8 (MUC2)
40	50	0.25%	0	63.1 (MUC2)
50	37	0.18%	0	37.2 (NBPF20)

Table 1: Spectral fibres of the 20 431 reviewed human proteins. N_k is the number of sequences with the same k -mer multiset.

quantiles are 7 and 20, and its maximum is 951 (NBPF20). The cumulative shares are:

$$\begin{aligned} \kappa \leq 4 : 10.0\%, \quad \kappa \leq 5 : 53.5\%, \quad \kappa \leq 6 : 83.1\%, \\ \kappa \leq 8 : 93.4\%, \quad \kappa \leq 10 : 95.8\%, \quad \kappa \leq 25 : 99.4\%. \end{aligned}$$

Separation is not resolution. Among the 20 353 distinct sequences, exactly one pair shares its amino acid composition. No two share their k -spectrum for $k = 2, 3, 4$. So even dipeptide composition separates every protein of the reference proteome from every other, and nothing in the *training set* of a protein encoder forces it to look further. What the reference set does not contain is the fibre: at $k = 3$ the median protein has 2^{239} spectral twins, and at $k = 4$ still 2^{19} . Those are exactly the inputs on which out-of-distribution generalization is tested: rearranged, recombined or unseen sequences. On them a local encoder of order 3 or 4 is constant by construction, whatever it learned. Separating a finite reference set and resolving sequence space are different properties, and benchmarks on the reference set test only the first.

Proteins named in UniPert-G2CP. Table 2 lists proteins that appear in the UniPert-G2CP analyses: the unseen single-gene targets CEP55, AURKA and AURKB, the combination CBL+CNN1, the kinesin/FOXM1 example, PD-1/PD-L1, the GABA_A receptor, and ESR1. All have κ between 5 and 11. The kinesin CENPE is the least resolved, with 226 bits of unresolved order at $k = 5$.

4 Where the unresolved order lives

Fibres grow with length at small k : a long protein repeats many short words by chance, so $\log_2 N_3/L$ is largest for titin (3.32 bits per residue), the mucins, nesprins and plakins. The spectral resolution does not grow with length in this way. It is large only if the protein contains long repeats

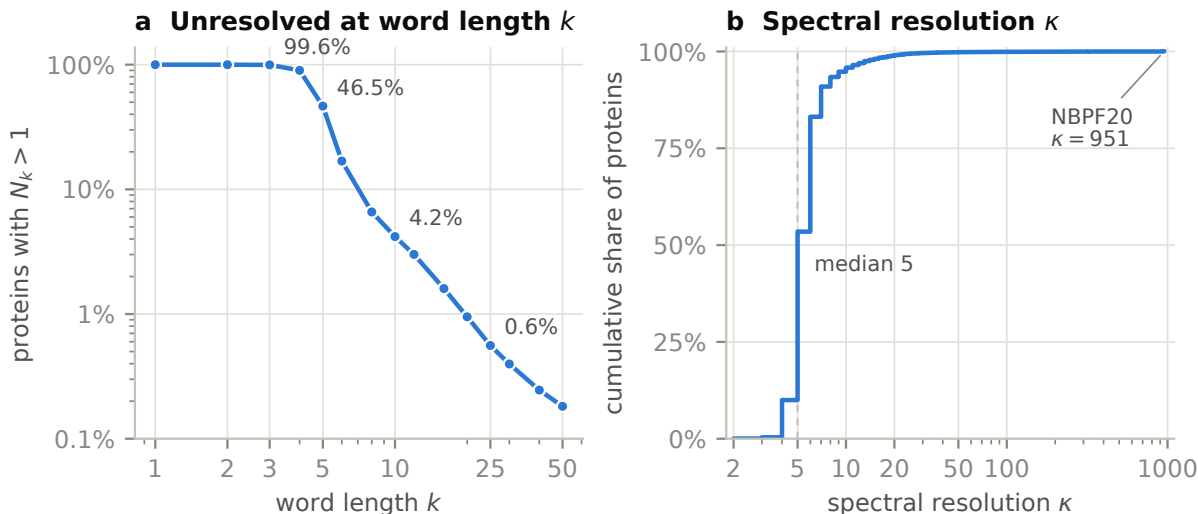


Figure 2: Spectral resolution of the 20 431 reviewed human proteins. (a) Share of proteins whose k -spectrum does not determine them ($N_k > 1$), on logarithmic axes. (b) Cumulative distribution of the spectral resolution κ . The dashed line marks the median, 5.

that interleave or recur, so it is the family-level statistic. Figure 3b shows this directly. Proteins of every length reach $\kappa \leq 10$, while the large values belong to particular families at moderate lengths.

Table 3 ranks families by median κ . The top of the list is not a random sample of repetitive proteins. The first three families and POTE arose by segmental duplication in the hominoid lineage. The NBPf proteins carry the Olduvai (DUF1220) domain, whose copy number expanded dramatically in humans [11], and NBPf20 has $\kappa = 951$. The next group consists of tandem-repeat families whose repeat units are identical or nearly identical: keratin-associated proteins, collagens, and the Krueppel C2H2 zinc-finger proteins. The latter are dominated by the KRAB zinc-finger proteins, which expanded with the transposable elements they control [12].

The zinc fingers are the clearest case. The 604 proteins whose family annotation contains “C2H2-type zinc-finger” are 3.0% of the proteome. Yet 348 of them have $N_{10} > 1$, i.e. 41% of the 854 proteins that remain unresolved at $k = 10$. ZNF91 and ZNF208, which have $\kappa = 51$ and 95, are typical. These are also the proteins whose *specificity* lies in which finger sits where: the DNA motif a KZFP recognizes is read off the order of its fingers. UniPert’s Fig. 2G singles out the zinc-finger family as one its embedding clusters well. That is consistent with our finding: a composition-type encoder sees these proteins as nearly interchangeable, while an alignment graph and a contextual language model do not.

5 Explicit collisions

For 114 human proteins $N_{25} > 1$, so PseAAC with $\lambda = 24$, the setting of the UniPert benchmark, is not injective on their fibres. To make this concrete, we take 12 of them spread over lengths 187–2078. For each, we draw a random Eulerian trail with the same start by Hierholzer’s algorithm, and keep it if it spells a different sequence. For every pair we then check exactly (in rational arithmetic) three things: the 25-spectra coincide; the 68-dimensional amphiphilic PseAAC vectors coincide; and the outputs of a random 8-channel integer convolutional network with window 25

protein	L	κ	$\log_2 N_3$	$\log_2 N_4$	$\log_2 N_5$	$\log_2 N_6$
ESR1	595	5	408.2	33.7	0	0
CEP55	464	6	361.9	43.5	1.6	0
AURKA	403	5	234.1	18.9	0	0
AURKB	344	5	152.2	6.6	0	0
CBL	906	7	878.8	90.8	15.2	1.6
CNN1	297	7	109.1	13.1	5.0	1.6
CENPE	2701	11	5298.0	1352.8	226.3	48.3
FOXM1	763	6	773.9	78.3	3.0	0
PDCD1 (PD-1)	288	5	128.3	9.2	0	0
CD274 (PD-L1)	290	6	92.0	6.2	1.0	0
GABRA1	456	5	265.1	19.4	0	0

Table 2: Proteins discussed in UniPert-G2CP [1]. $\log_2 N_k$ is the number of bits of order that a local encoder of order k cannot see.

family (UniProt)	members	median κ	origin of the repeats
NBPF	16	46	Olduvai (DUF1220) domain amplification, hominoid-specific
GOLGA6	17	19	segmental duplication, hominoid-specific
NPIP	20	16	segmental duplication, hominoid-specific
KRTAP type 4	12	14	cysteine-rich pentapeptide repeats
Krueppel C2H2 zinc finger	536	13	tandem C2H2 fingers
KRTAP types 5, 10	11, 13	13	cysteine-rich repeats
POTE	13	12	segmental duplication, primate-specific
KRTAP types 28, 9	10, 8	11, 10	cysteine-rich repeats
collagens (fibrillar, FACIT)	11, 9	10	Gly-X-Y triplets
all proteins	20 431	5	

Table 3: Families with at least eight members, ranked by median spectral resolution.

and mean pooling coincide. The PseAAC vectors were computed with the Kyte–Doolittle and Hopp–Woods scales, but by Proposition 2 the scales are irrelevant. The pairs are far from identical (Table 4). Figure 4 shows where they differ. The changes come in contiguous blocks, the transposed segments, and in the zinc-finger proteins these blocks span several fingers.

The pairs changed at two positions deserve a remark. There, two copies of a repeat unit differ at one residue each, and the fibre-mate swaps the two variants between the copies. A local encoder of order 25 cannot tell *which copy carries which variant*. In the zinc-finger pairs, whole fingers change places (10–15% of positions). The alignment score drops to 0.77–0.85 of the self-score, so the two sequences are far apart for an alignment-based encoder, and yet they are indistinguishable for PseAAC.

6 The chemical side: ECFP

ECFP is the molecular analogue of a k -mer spectrum [2]. After r rounds, each atom carries an identifier of its radius- r environment. The fingerprint is the *set* of identifiers of all atoms, hashed

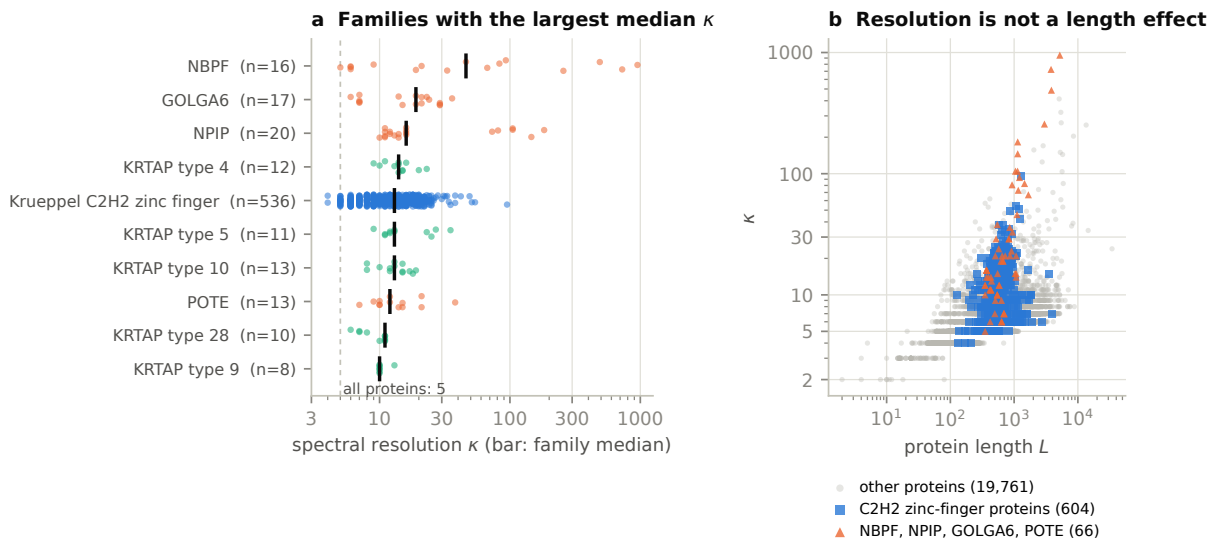


Figure 3: Where the unresolved order lives. (a) The ten families (at least eight members) with the largest median κ . Each dot is a protein, and the bar is the family median. Orange: hominoid/primate-specific segmental-duplication families; blue: Krueppel C2H2 zinc-finger proteins; green: keratin-associated proteins. The dashed line is the proteome median. (b) κ against protein length for all proteins. The two highlighted groups are shown in their own colours and marker shapes.

into 2048 bits, with multiplicities discarded in the binary form that UniPert uses ($r = 2$). It is therefore local in the graph sense. Molecules with the same radius-2 environment set receive the same fingerprint. The refinement is the Weisfeiler–Lehman colour refinement, whose blind spots (regular and symmetric graphs, rings of equal local structure) are well documented [13]. Binarization makes the fibre larger than its sequence counterpart, since it forgets how many atoms share an environment. The radius also fixes a horizon. A radius-2 environment reaches two bonds from its centre, so no single identifier spans more than four bonds. Two molecules that assemble the same environments into different long-range arrangements (the order of substituents along a chain, the placement of equal units around a macrocycle) therefore lie in the same fibre. This is the exact analogue of a transposition between repeated $(k - 1)$ -mers. The default ECFP also carries no stereochemistry, so stereoisomers (enantiomers, *cis/trans* isomers) always share a fingerprint unless chirality flags are switched on. Both are *molecular fibres* in the sense of this paper: sets of distinct perturbagens that a local encoder cannot tell apart, whatever it was trained on. We do not compute molecular fibres here. We note only that in UniPert-G2CP both perturbagen types would have been local, had the protein encoder been PseAAC: the gain on the genetic side comes precisely from giving up locality.

7 DNA perturbagens and the double cover

UniPert-G2CP names as its limitation that genetic perturbations outside protein-coding genes cannot be represented, and points to DNA language models. DNA brings one more symmetry. A regulatory element, or a CRISPR target, has no preferred strand, so a DNA encoder that is to be strand-agnostic must satisfy $\phi(S) = \phi(\bar{S})$ for the reverse complement $\bar{S}$. A local encoder with this

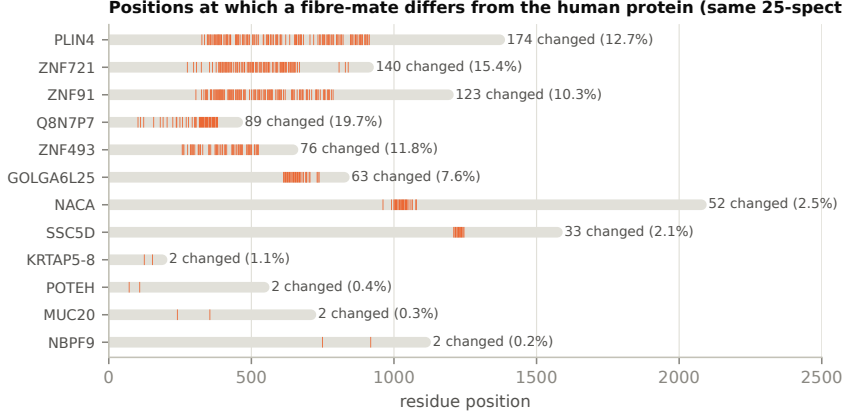


Figure 4: The twelve fibre-mates of Table 4. Each grey bar is a human protein, and orange marks the positions at which its fibre-mate differs. Each pair has the same 25-spectrum, the identical 68-dimensional PseAAC vector and the identical output from a window-25 convolutional encoder.

property depends only on the *double-stranded* spectrum $\text{spec}_k(S) + \text{spec}_k(\bar{S})$. Its fibre is the set $\text{DS}(S, k)$ of double-stranded reconstructions.

Proposition 7. *A strand-agnostic local encoder of order $k + 1$ is constant on $\text{DS}(S, k)$. For circular S ,*

$$|\text{DS}(S, k)| = \sum_f N_{\text{seq}}\left(\frac{m+f}{2}\right),$$

the sum running over the points f of the (-1) -eigenlattice of the reverse-complement involution on the circulations of the double cover D_k , inside the box $|f| \leq m$ [15, Theorem 7.6, p. 52].

As in [15, Definition 7.3, p. 51], $\text{DS}(S, k)$ is defined by the $(k + 1)$ -mer spectra of both strands, the k -mers being the vertices of D_k . An encoder of order $k + 1$ therefore sees exactly the double-stranded data defining $\text{DS}(S, k)$, and an encoder of any smaller order sees less, so it is constant on $\text{DS}(S, k)$ as well.

This fibre is not given by a single determinant. It is a lattice sum of determinants, and it is much harder to compute: counting double-stranded reconstructions is $\#P$ -hard in general [18, Corollary 10.6, p. 78], so no closed formula exists unless $\text{FP} = \#P$. For the 5386-bp genome of ϕX174 at $k = 10$, the lattice has rank 48 and the box has 2 481 687 900 points, of which 909 103 210 have connected support [16, Section 8.5 and Table 8.2, p. 63]. The number of double-stranded reconstructions is

$$|\text{DS}(\phi\text{X174}, 10)| = 31\,925\,753\,246\,212$$

[17, Section 9.8 and Table 9.1, pp. 72–73]. A strand-agnostic local DNA encoder of window 11, or of any smaller window, assigns all of them the same vector. The design choice “make the encoder reverse-complement invariant” is sound, but it has a price. For local encoders that price is exactly the passage from a sequence fibre to a double-stranded fibre.

The rearrangements hidden in that passage can be named. On one strand, the members of a fibre differ by transpositions of segments between direct repeats [8, 9]. On two strands there is a second move: a segment flanked by an inverted repeat $w \cdots \bar{w}$, with w a k -mer, may be replaced by its reverse complement without changing the double-stranded $(k + 1)$ -mer spectrum [19, Theorem 11.1, p. 84]. Transpositions between direct repeats and inversions between inverted

protein	L	positions changed	$SW(S, T)/SW(S, S)$	PseAAC, CNN
KRTAP5-8	187	2 (1.1%)	0.984	identical
Q8N7P7	452	89 (19.7%)	0.814	identical
POTEH	545	2 (0.4%)	0.994	identical
ZNF493	646	76 (11.8%)	0.824	identical
MUC20	709	2 (0.3%)	0.996	identical
GOLGA6L25	826	63 (7.6%)	0.944	identical
ZNF721	911	140 (15.4%)	0.769	identical
NBPF9	1111	2 (0.2%)	0.997	identical
ZNF91	1191	123 (10.3%)	0.846	identical
PLIN4	1371	174 (12.7%)	0.815	identical
SSC5D	1573	33 (2.1%)	0.981	identical
NACA	2078	52 (2.5%)	0.986	identical

Table 4: A human protein S and a different sequence T with the same 25-spectrum and the same start. SW is a Smith–Waterman local alignment score (match 2, mismatch -1 , gap -2). PseAAC is the 68-dimensional amphiphilic form with $\lambda = 24$; CNN is an 8-channel window-25 integer network with mean pooling.

repeats are the two classical products of non-allelic homologous recombination, so a strand-agnostic local DNA encoder of window $k + 1$ is blind to exactly the rearrangements that recombination between repeats of length k produces. For ϕ X174 at $k = 11$ and $k = 12$ these two moves, together with global reverse complementation, connect all 864 and 10 double-stranded reconstructions; that they connect every double-stranded fibre is a conjecture [19, Section 11.4 and Conjecture 11.2, pp. 85–86].

8 What this does and does not imply

Table 5 sets our results against the statements of UniPert-G2CP. Three limitations should be stated plainly.

First, a fibre is a set of *sequences*, not of proteins. Most fibre members are not biologically plausible, and a model trained on real proteins may never meet them. The fibres that matter are those reached by real processes, and here one must distinguish. Exon skipping, exon duplication and domain deletion change the k -mer content: they add or remove k -mers. So they move a sequence *out of* its fibre, and a local encoder does register them. What a local encoder cannot register is a change of order at constant content. Such changes include: gene conversion or unequal crossing-over that swaps variants between repeat copies; a variant assigned to the wrong copy of a repeat; and a reordering of identical or near-identical units, such as fingers in a KZFP, Olduvai domains in an NBPF protein, or cysteine-rich repeats in a KRTAP. These are the transpositions of Section 2, and they are concentrated in the families of Table 3. Whether such events have phenotypic consequences that a perturbation model should predict is open. When they do, a local encoder cannot predict them.

Second, we have not evaluated UniPert, ESM or any trained model. Our statements about PseAAC and convolutional encoders are theorems about their form. The comparison with UniPert’s benchmark offers a mechanism consistent with their ranking, not a test of it. Such a test is easy to set up: a *fibre-mate benchmark*. It would pair each protein with a sequence from its k -fibre (for example, a repeat-copy swap), and ask whether an encoder separates the pair and whether a

UniPert-G2CP	what the fibre analysis adds
UniPert outperforms PseAAC, ESM, OntoProtein in genetic OOD tasks	PseAAC ($\lambda = 24$) is constant on 25-fibres: provably identical vectors for 12 explicit human protein pairs differing at up to 15% of positions; 114 proteins have non-trivial 25-fibres
sequence-based encoders “risk losing local information”	composition-type encoders lose <i>global order</i> , not local content: median 2^{239} twins at $k = 3$, 2^{19} at $k = 4$; median resolution $\kappa = 5$
reference set of 19 187 proteins	every local encoder of order ≥ 2 already separates the reference proteome; separation of the reference set is not evidence of resolution
zinc-finger family clustered well (Fig. 2G)	KZFPs are 3% of proteins but 41% of those with $\kappa > 10$; their function is in finger order, which local encoders cannot see
ESR1 domains and mutants (Fig. 6)	$\kappa(\text{ESR1}) = 5$; a point substitution changes at most k of the k -mers, and it is visible to any local encoder of order $\leq \kappa$
ECFP4 for compounds	ECFP is a local (radius-2, binarized) encoder; its blind spots are those of Weisfeiler–Lehman refinement
limitation: non-coding perturbations	a strand-agnostic local DNA encoder is constant on the double-stranded reconstruction set, $3.19 \cdot 10^{13}$ sequences for ϕX174 at $k = 10$

Table 5: The results of this paper set against the corresponding statements of UniPert-G2CP.

predictor assigns them different effects.

Third, the spectral resolution measures what is *invisible* to local statistics, not what is *relevant* to a phenotype. Whether finger order in a KZFP changes a perturbation phenotype is a biological question. Our analysis says only that a local encoder cannot represent the answer.

Data and code availability

`spectral_fibres.py` (Python 3.10+, standard library; `numpy` only for the floating-point determinants) downloads the reviewed human proteome from UniProt. It then runs the five checks (formula, contraction, proteome, collisions, fibre-mates) and writes the per-protein results (`fibre_results.json`). The transcript is `spectral_fibres_output.txt`. The script, its transcript and the figure script are in the repository of [14], <https://github.com/Ruqing1963/algebraic-genomics>, folder `papers/Ch12_Spectral_Fibres`. The double-stranded count of Section 7 is reproduced by `ds_fermion.py` and `ds_count.py`, the scripts of [17], in the same repository.

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